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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 15

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	2025 – 2026, Semester VII
Date of Performance	23rd July, 2026
Title of Assignment:	Basic Infrastructure as Code (IaC) with Terraform

Theory:	Infrastructure as Code (IaC) is the practice of managing and provisioning infrastructure using configuration files instead of manual processes. Terraform is an open-source IaC tool developed by HashiCorp that enables users to define, provision, modify, and destroy infrastructure resources in a consistent and automated manner. In this experiment, Terraform was used with the Local provider to create, update, and destroy a local text file. The commands terraform init, terraform plan, terraform apply, and terraform destroy demonstrated Terraform's workflow of initializing the working directory, previewing infrastructure changes, applying the desired configuration, and safely removing managed resources. This approach ensures repeatable deployments, version control of infrastructure, and simplified infrastructure management. **Commands Used** • terraform --version – Checks whether Terraform is installed correctly and displays the installed version. • mkdir my-terraform-iac – Creates a new directory for storing the Terraform project files. • cd my-terraform-iac – Navigates to the Terraform project directory. • terraform init – Initializes the Terraform working directory by downloading the required providers and preparing the project. • terraform plan – Generates an execution plan to preview the infrastructure changes before applying them. • terraform apply – Applies the Terraform configuration and creates or

	updates the resources defined in the configuration file. · terraform destroy – Removes all resources managed by the current Terraform configuration.

Implementation:

Step 1: Install Terraform

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % brew --version
Homebrew 6.0.22
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % brew tap hashicorp/tap
brew install hashicorp/tap/terraform
==> Auto-updating Homebrew...
Adjust how often this is run with `$HOMEBREW_AUTO_UPDATE_SECS` or disable with
`$HOMEBREW_NO_AUTO_UPDATE=1`. Hide these hints with `$HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
==> Downloading https://ghcr.io/v2/homebrew/core/portable-ruby/blobs/sha256:25d6becc4464b6e65ac38a76ccd27eee8be5d0d805f6b3292d74e37
==> Pouring portable-ruby-4.0.7_1.arm64_big_sur.bottle.tar.gz
==> Auto-updated Homebrew!
Updated 4 taps (cloudfoundry/tap, mongodb/brew, homebrew/core and homebrew/cask).
==> New Formulae
agent-manager: Run Claude Code, Codex, OpenCode and other AI coding agents in tmux
all-smi: GPU monitoring tool for NVIDIA/Jetson/Apple Silicon/Tenstorrent
android-file-transfer-linux: Command-line MTP client for Android and other MTP devices
annotsv: Annotation and ranking of structural variations
apkid: Android application identifier for compilers, packers, and obfuscators
arcane-cli: Command-line client for the Arcane Docker management platform
ascelerate: Command-line tool for the App Store Connect API
bashka: Static verification of installation bash scripts
baysor: Bayesian segmentation of imaging-based spatial transcriptomics data
bend: Language that blocks AI mistakes via proof
bitbucket-cli: CLI for Bitbucket Cloud and Data Center
bqlog: Cross-language logging library
buildkite: Work with Buildkite from the command-line
ccmux: Run all your AI coding agents in tmux
cellpose: Generalist algorithm for cell and nucleus segmentation
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % terraform --version
Terraform v1.17.0
on darwin_arm64
```

Installed Terraform — Successfully installed Terraform on the MacOS system and verified the installation using the terraform --version command.

Step 2: Create the Project Directory

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % cd "/Users/pushkrajpnaik/Desktop/Pushkraj /DevOps/DevOps-Assignments/Assignment-15-Terraform-IaC"
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % terraform init
Terraform initialized in an empty directory!

The directory has no Terraform configuration files. You may begin working
with Terraform immediately by creating Terraform configuration files.
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC %
```

Created Terraform Project Directory — Created a new project directory named my-terraform-iac and navigated into it to store the Terraform configuration files.

Step 3: Create the Terraform Configuration File (main.tf)

DevOps-Assignments > Assignment-12-Kubernetes-Deployment > k8s-manifests > ! web-deployment.yaml > apiVersion

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-app-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: web-app
  template:
    metadata:
      labels:
        app: web-app
    spec:
      containers:
        - name: web-app
          image: my-flask-app:v2
          imagePullPolicy: IfNotPresent
          ports:
            - containerPort: 5001

```

Created Terraform Configuration File — Created the main.tf configuration file and defined the local provider along with a local_file resource to create a text file using Terraform.

Step 4: Initialize Terraform

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/local versions matching "~> 2.5"...
- Installing hashicorp/local v2.9.1...
- Installed hashicorp/local v2.9.1 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC %
```

Initialized Terraform — Initialized the Terraform working directory using the terraform init command, which downloaded the required provider plugins and prepared the project for execution.

Step 5: Plan the Changes

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # local_file.example will be created
  + resource "local_file" "example" {
      + content              = "Hello from Terraform Infrastructure as Code!"
      + content_base64sha256 = (known after apply)
      + content_base64sha512 = (known after apply)
      + content_md5          = (known after apply)
      + content_sha1         = (known after apply)
      + content_sha256       = (known after apply)
      + content_sha512       = (known after apply)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "./example.txt"
      + id                   = (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC %
```

Generated Terraform Execution Plan — Generated an execution plan using the terraform plan command to preview the changes Terraform would make. The plan indicated that one new resource (local_file.my_example_file) would be created with no modifications or deletions.

Step 6: Apply the Changes

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # local_file.example will be created
  + resource "local_file" "example" {
      + content              = "Hello from Terraform Infrastructure as Code!"
      + content_base64sha256 = (known after apply)
      + content_base64sha512 = (known after apply)
      + content_md5          = (known after apply)
      + content_sha1         = (known after apply)
      + content_sha256       = (known after apply)
      + content_sha512       = (known after apply)
      + directory_permission = "0777"
      + file_permission      = "0777"
      + filename             = "./example.txt"
      + id                   = (known after apply)
    }

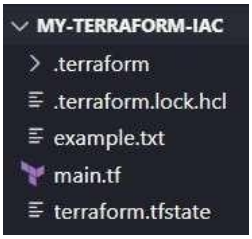
Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.example: Creating...
local_file.example: Creation complete after 0s [id=934365396dcb63e6fbf85a11dd852ae991ac34c2]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```



Applied Terraform Configuration — Applied the Terraform configuration using the terraform apply command, which successfully created the example.txt file as defined in the main.tf configuration. The generated file was verified in the project directory.

Step 7: Modify the Content and Re-apply

Command: terraform plan

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % terraform plan
local_file.example: Refreshing state... [id=934365396dcb63e6fbf85a11dd852ae991ac34c2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # local_file.example must be replaced
-/+ resource "local_file" "example" {
      ~ content              = "Hello from Terraform Infrastructure as Code!" -> "Terraform successfully updated this file!" # forces replacement
      ~ content_base64sha256 = "1GgWWBZQ9gXvi3wBYBc26DzuWlLMH1CTlh9ZEjzxxJo=" -> (known after apply)
      ~ content_base64sha512 = "kofuqnCyBFiUxJH7EAeL4318qmmu05D0AxGf5SjB3WiXVzgEUT4MAkTaWj2iO6tf9JrJaybb545P0YQgc3wjJQ==" -> (known after apply)
      ~ content_md5          = "330cee5c064fbce2469f7e5a9eab8a40" -> (known after apply)
      ~ content_sha1         = "934365396dcb63e6fbf85a11dd852ae991ac34c2" -> (known after apply)
      ~ content_sha256       = "d46816581650f605ef8b7c01601736e83cee5a52cc1f5093961f59123cf1c49a" -> (known after apply)
      ~ content_sha512       = "9287eeaa70b2045894c491fb10078be37d7caa69aed390f403119fe528c1dd6897573804513e0c0244da5a3da23bab5ff49ac96b26dbe78e4fd1842073
7c2325" -> (known after apply)
      ~ id                   = "934365396dcb63e6fbf85a11dd852ae991ac34c2" -> (known after apply)
        # (3 unchanged attributes hidden)
    }

Plan: 1 to add, 0 to change, 1 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.
```

Generated Updated Execution Plan — Modified the main.tf configuration and generated an updated execution plan using the terraform plan command. Terraform detected that the existing local_file resource would be replaced to apply the updated content.

Command: terraform apply

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % terraform apply
local_file.example: Refreshing state... [id=934365396dcb63e6fbf85a11dd852ae991ac34c2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement

Terraform will perform the following actions:

  # local_file.example must be replaced
-/+ resource "local_file" "example" {
      ~ content              = "Hello from Terraform Infrastructure as Code!" -> "Terraform successfully updated this file!" # forces replacement
      ~ content_base64sha256 = "1GgWWBZQ9gXvi3wBYBc26DzuWlLMH1CTlh9ZEjzxxJo=" -> (known after apply)
      ~ content_base64sha512 = "kofuqnCyBFiUxJH7EAeL4318qmmu05D0AxGf5SjB3WiXVzgEUT4MAkTaWj2iO6tf9JrJaybb545P0YQgc3wjJQ==" -> (known after apply)
      ~ content_md5          = "330cee5c064fbce2469f7e5a9eab8a40" -> (known after apply)
      ~ content_sha1         = "934365396dcb63e6fbf85a11dd852ae991ac34c2" -> (known after apply)
      ~ content_sha256       = "d46816581650f605ef8b7c01601736e83cee5a52cc1f5093961f59123cf1c49a" -> (known after apply)
      ~ content_sha512       = "9287eeaa70b2045894c491fb10078be37d7caa69aed390f403119fe528c1dd6897573804513e0c0244da5a3da23bab5ff49ac96b26dbe78e4fd184
7c2325" -> (known after apply)
      ~ id                   = "934365396dcb63e6fbf85a11dd852ae991ac34c2" -> (known after apply)
        # (3 unchanged attributes hidden)
    }

Plan: 1 to add, 0 to change, 1 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

local_file.example: Destroying... [id=934365396dcb63e6fbf85a11dd852ae991ac34c2]
local_file.example: Destruction complete after 0s
local_file.example: Creating...
local_file.example: Creation complete after 0s [id=c0eb713098a9a6b17dd97ac8a3e492f8704d04b5]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
```

Applied Updated Terraform Configuration — Applied the modified Terraform configuration using the terraform apply command. Terraform successfully recreated the local_file resource with the updated content.



Verified Updated File Content — Verified that the example.txt file was successfully updated with the new content after applying the modified Terraform configuration.

Step 8: Destroy the Resource

Command: terraform destroy

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % terraform destroy
local_file.example: Refreshing state... [id=c0eb713098a9a6b17dd97ac8a3e492f8704d04b5]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
  - destroy

Terraform will perform the following actions:

  # local_file.example will be destroyed
  - resource "local_file" "example" {
      - content              = "Terraform successfully updated this file!" -> null
      - content_base64sha256 = "5cqMtlCeHzHfkxLfEpa10RbF6+6HjlFQLN0pOpGxVcA=" -> null
      - content_base64sha512 = "o1zMjYp6qUo4JlHoPPSoluJrLmxsiVR/LRpUoVYxMWpDaGWH0jUX5DQFW0x0qQB5l6qTbyZRB2Wc3cc6c0tmdQ==" -> null
      - content_md5          = "3f089cbcbbfe3d3f584dc5432b36c866" -> null
      - content_sha1         = "c0eb713098a9a6b17dd97ac8a3e492f8704d04b5" -> null
      - content_sha256       = "e5ca8cb6509e1f31df9312df1296b5d116c5ebee878e51502cdd293a91b155c0" -> null
      - content_sha512       = "a35ccc8d8a7aa94a382651e83cf4a896e26b2e6c6c89547f2d1a54a15631316a43686587d23517e434055b4c74a900799
4b6675" -> null
      - directory_permission = "0777" -> null
      - file_permission      = "0777" -> null
      - filename             = "./example.txt" -> null
      - id                   = "c0eb713098a9a6b17dd97ac8a3e492f8704d04b5" -> null
    }

Plan: 0 to add, 0 to change, 1 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes

local_file.example: Destroying... [id=c0eb713098a9a6b17dd97ac8a3e492f8704d04b5]
local_file.example: Destruction complete after 0s

Destroy complete! Resources: 1 destroyed.
```

Destroyed Terraform Resource — Executed the terraform destroy command, confirmed the destruction by entering yes, and successfully removed the Terraform-managed local_file resource from the local system.

```
(base) pushkrajpnaik@Pushkrajs-Laptop Assignment-15-Terraform-IaC % ls -la
total 32
drwxr-xr-x   7 pushkrajpnaik  staff   224 Oct 26 04:46 .
drwxr-xr-x  19 pushkrajpnaik  staff   608 Oct 26 04:40 ..
drwxr-xr-x   3 pushkrajpnaik  staff    96 Oct 26 04:43 .terraform
-rw-r--r--   1 pushkrajpnaik  staff  1257 Oct 26 04:43 .terraform.lock.hcl
-rw-r--r--   1 pushkrajpnaik  staff   256 Oct 26 04:45 main.tf
-rw-r--r--   1 pushkrajpnaik  staff   181 Oct 26 04:46 terraform.tfstate
-rw-r--r--   1 pushkrajpnaik  staff  1658 Oct 26 04:46 terraform.tfstate.backup
```

Verified Resource Deletion — Verified that the example.txt file was successfully removed from the project directory after the Terraform-managed resource was destroyed.

Subject Faculty Name: Prof. Arijit Dutta